

Evaluation Goals for Online Process Mining: a Concept Drift Perspective

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Online process mining refers to a class of techniques for analyzing in real-time event streams generated by the execution of business processes. These techniques are crucial in the reactive monitoring of business processes, timely resource allocation and detection/prevention of dysfunctional behavior. Many interesting advances have been made by the research community in recent years, but there is no consensus on the exact set of properties these techniques have to achieve. This

paper fills the gap by identifying a set of evaluation goals for online process mining and examining their fulfillment in the state of the art. We discuss parameters and techniques regulating the balance between conflicting goals and outline research needed for their improvement. Concept drift detection is crucial in this sense but, as demonstrated by our experiments, it is only partially supported by current solutions.